Malware - Overview

February 3, 2026

Université Perpignan Via Domitia

Outline

1. Introduction - Taxonomy
   - Definition and Classification
   - Logic Bomb
   - Trojan Horses
   - Viruses and Worms
2. Virus Fundamentals
3. Virus Infection Methods
4. Virus and Worm Classification
5. Conclusion

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Definition of Malware

Definition (Malware):

A *simple program* or *self-replicating program*:

- with offensive intent, installing itself in an information system without the user's knowledge,
- designed to compromise the confidentiality, integrity, or availability of that system,
- or capable of falsely incriminating its owner or user in criminal activity.

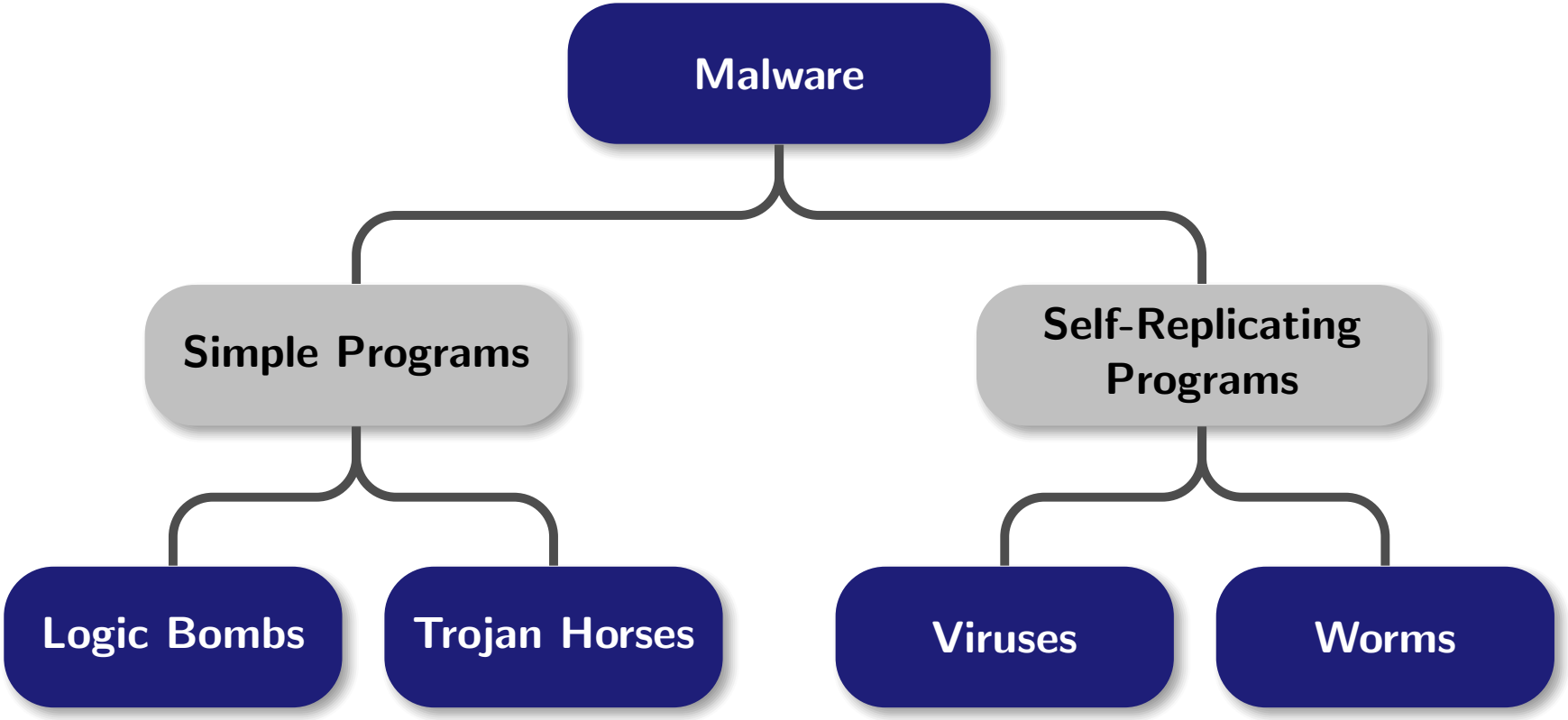
Postulate

Malware can affect any environment capable of code execution!

- All operating systems.
- Mobile environments (smartphones, gaming consoles, GPS devices, embedded computers. . .).
- Many document formats.

Adleman's Classification

Adleman classified malicious software into 4 categories.



The main distinction is:

- *Simple program*: a standard program with malicious intent.
- *Self-replicating*: a program that copies its own code, typically into other executables.

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Logic Bomb

Definition (Logic Bomb)

A logic bomb is a simple malicious program that installs itself in a system and waits for a specific event (date, action, particular data...) called a trigger to execute its offensive function.

These programs often constitute the payload of a virus (e.g., the CIH virus that activates every April 26th).

Logic Bomb Example

- A system administrator had implanted a program that checked for the presence of his name in the company's payroll records.
- If his name was absent (indicating termination), the program would encrypt all company hard drives.

- Without the encryption key, the company could no longer access its data.

Limitations of Antivirus Protection Against Logic Bombs

- Antivirus software struggles to detect logic bombs, as they appear to be ordinary programs.
- Advanced antivirus techniques (heuristic analysis, code emulation) are inherently limited against unknown logic bombs.

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Trojan Horses

Definition (Trojan Horse)

A Trojan horse is a simple program composed of two parts:

- *the server module:*
  - *Installed on the victim's computer.*
  - *Covertly provides the attacker with access to all or part of the system resources over the network.*
- *the client module:*
  - *Located on the attacker's machine.*
  - *Allows access to services unknowingly provided by the victim.*

Trojan Horses

The server module:

- Usually hidden within another *harmless and attractive program.*
- Executing this program installs the Trojan's server component without the victim's knowledge.

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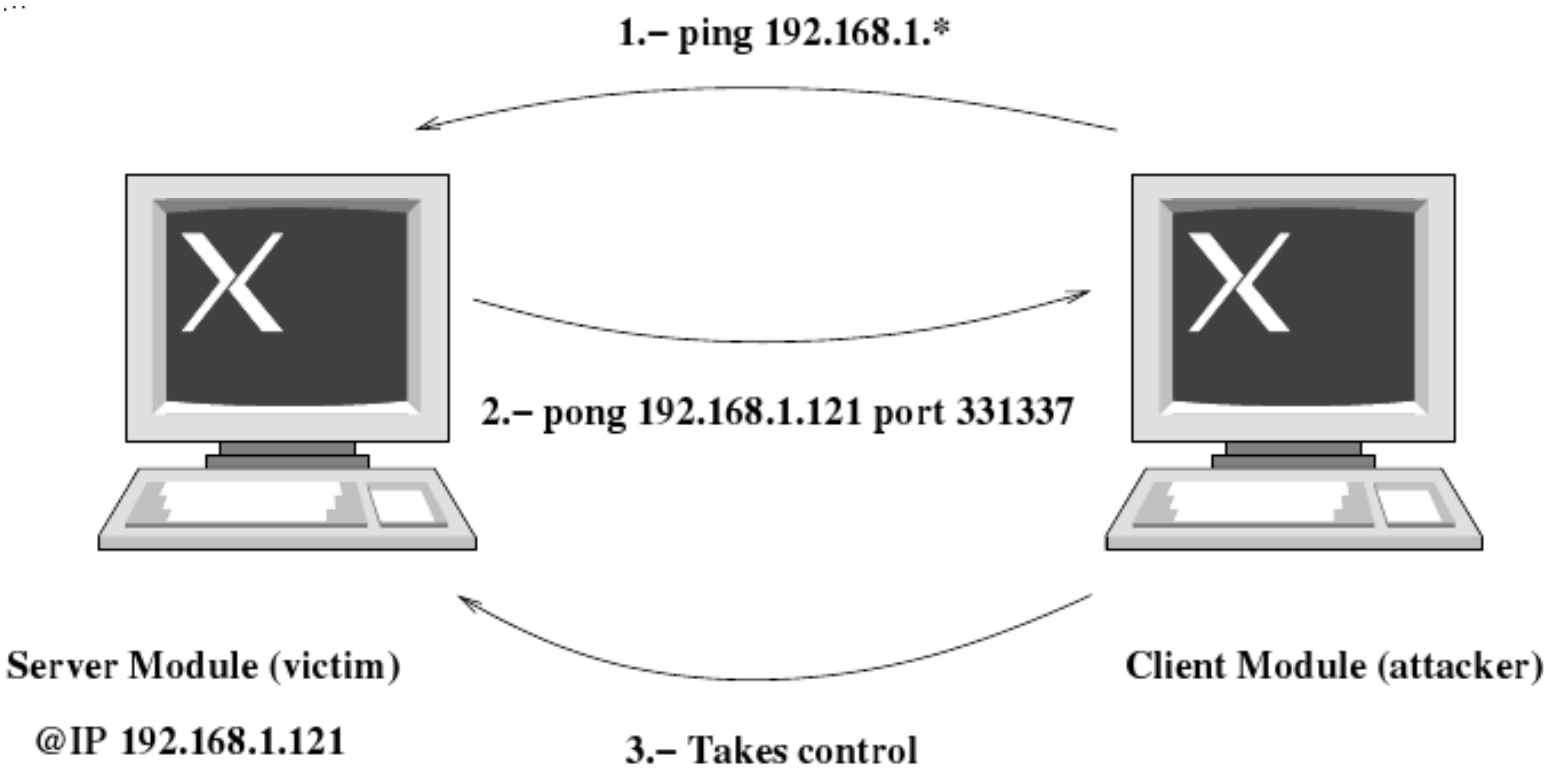
The client module:

- Scans the network for IP addresses and ports (TCP or UDP) of machines infected by the server module.
- Takes control of the infected machines found.

Client-Server Architecture in Trojan Horses



Trojan Horse Detection

Detection of these programs is inconsistent.

- A well-crafted Trojan horse not distributed on the internet has a high chance of evading antivirus detection.
- Using a firewall is highly recommended, although several techniques exist or are being developed to bypass them.

Trojan Horse Examples

- Decoys: programs mimicking the normal behavior of a legitimate system program (e.g., fake Unix login banner).
- Keyboard spies, known as *keyloggers*, are a specific type of Trojan horse.

The offensive action typically involves stealing one or more pieces of information from the victim's machine.

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Worms and Viruses: Definitions

Definition (Virus)

A program that propagates by copying its own code.

Definition (Worm)

Can be viewed as a network-oriented virus. The main difference from a virus is that a worm is not necessarily attached to a file (e.g., simple worms like Slammer).

Computer Worms

Three main classes of worms:

- Simple worms. Exploit security vulnerabilities (Slammer, Sasser,. . .).
- Macro worms. Use social engineering and infected office document attachments.
- Email worms. Use social engineering and infected executable attachments (Bagle, Netsky,. . .).

In the following sections, we will focus primarily on self-replicating programs:

- We will mainly study viruses,
- and to a lesser extent, worms.

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   - Statistics and Metrics
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Propagation Method

The malicious program is carried by a host program or *dropper*.

When the *dropper* is executed:

1. The malicious program (or viral code) takes control:
   - It operates according to its own mode.
   - The host program is temporarily suspended.
2. It returns control to the host program, which then executes normally without revealing the presence of the malicious code.

Functional Diagram of a Virus

1. A *search* routine to find target programs.
2. A *copy* routine: injection of viral code into the target.
3. An *anti-detection* routine. Purpose: to counter antivirus action.
4. The *payload*, possibly coupled with a delayed activation mechanism (*trigger*).

Life Cycle of a Virus or Worm

Three phases can be identified in the life of a virus:

- The infection phase.
- The dormant phase (incubation).
- The execution phase (payload activation).

Infection Phase

During this phase, the virus spreads through the environment:

- *Passively*
  - The infected medium is copied to a storage device (CD-ROM, download site, forum,. . .) and transmitted.
  - Victims can then copy and execute it in their own environment.
- *Actively*
  - The user executes the dropper or a file already contaminated during a previous infection.

Infection Phase

During this phase, the virus searches for target programs:

- An effective virus will verify that the file is executable, has the correct format, and that the target is not already infected.
- It also ensures to minimize detection risks: for example, infection of *.COM executables must not exceed 64KB (maximum size of *.COM files).

The virus then copies its own code into the target.

Dormant Phase (Incubation)

- This is the longest part of a virus's life.
- Primary mission: ensure the virus's survival.
- The goal is to limit or prevent its detection:
  - Detection by the user: by avoiding, for example, execution errors.
  - Detection by antivirus: evading antivirus surveillance.

Execution Phase (Payload Activation)

During this phase, *the payload is activated*. Its triggering depends on its position:

- *At the beginning of the code*: the payload will be executed before any infection.
- *At the end of the code*: it will only occur after the infection process.
- *In the middle of the code*: particularly if it is conditioned by the success or failure of the infection.

Execution Phase: Triggering

- Triggering can be delayed (trigger mechanism).
- The payload is then a logic bomb mounted on a viral vector.

Examples:

- System date.
- Number of infections completed.
- A particular key sequence pressed a certain number of times (e.g., `CTRL+ALT+DEL`).
- Number of times a `Word` document is opened.

Execution Phase: Payload Effects

The effects of the payload are highly variable:

- *Non-destructive*: displaying images or animations, messages, ...
- *Destructive*: the aim is to compromise
  - Data confidentiality (data exfiltration).
  - System or data integrity (file destruction/modification, ...)
  - System availability (CPU saturation, ...)
  - Data availability (encryption) or user incrimination.

Execution Phase: Hardware Destruction?

- Generally not accepted by experts.
- CIH controversy:
  - CIH does not directly destroy hardware but the associated software (firmware, etc.).
  - The hardware attack is only simulated.
- Other older examples: attacking floppy or hard disk drives through repeated I/O requests.

Biological/Computer Comparison

Biological Virus	Computer Virus
Specific cell attacks by format	Specific format attacks
Infected cells produce new viruses	Infected program generates other viral programs
Modification of cell's genetic information	Modification of program actions
Virus uses host cell structures to multiply	Multiplication only via infected program
Viral interactions Multiplication only in living cells	Binary viruses or anti-antivirus Execution required for dissemination
An infected cell is not reinfected by the same virus	Reinfection prevention
Retrovirus	Virus specifically targeting an antivirus - Source code virus
Viral mutation	Viral polymorphism
Healthy carrier	Latent viruses
Antigens	Signatures

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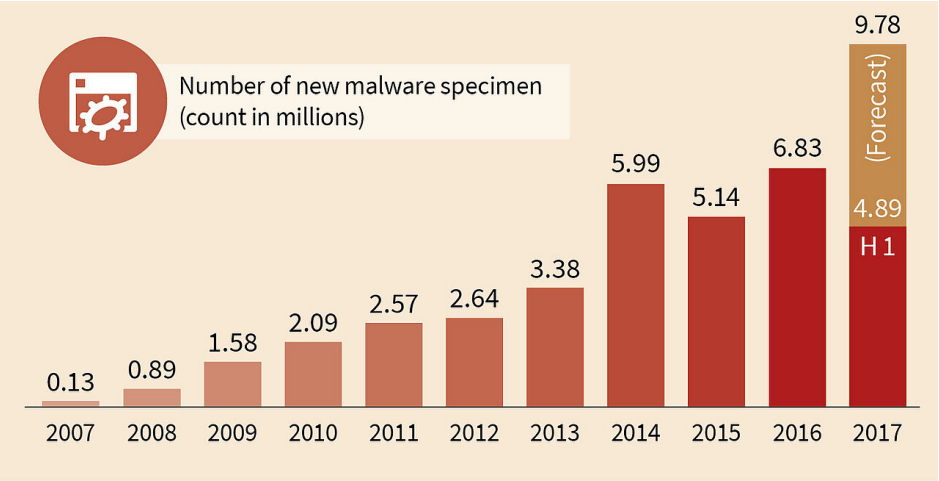
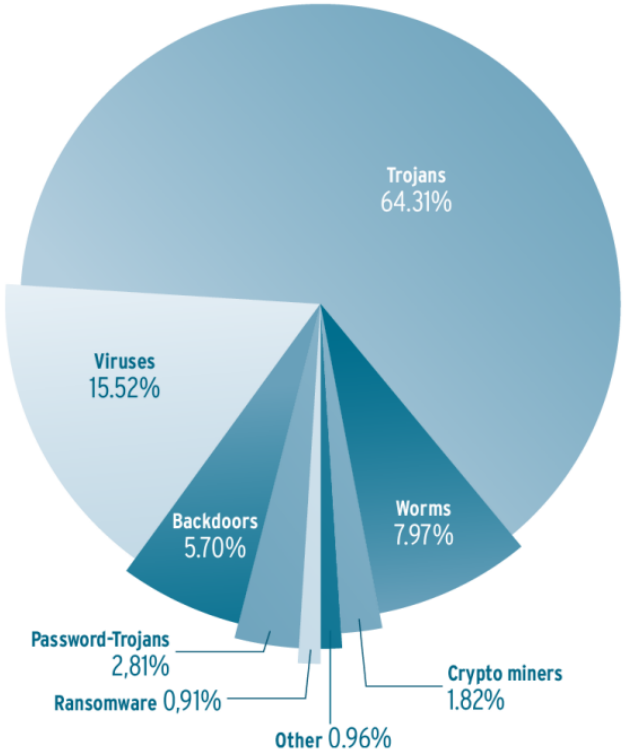
Some Statistics

Statistics on malware are difficult to obtain and verify (...)





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Overwriting Viruses

When the virus executes:

- The search routine identifies targets.
- Infection occurs by simply overwriting the target code with its own code.
- These viruses generally lack a payload, as the infection method itself serves as one.

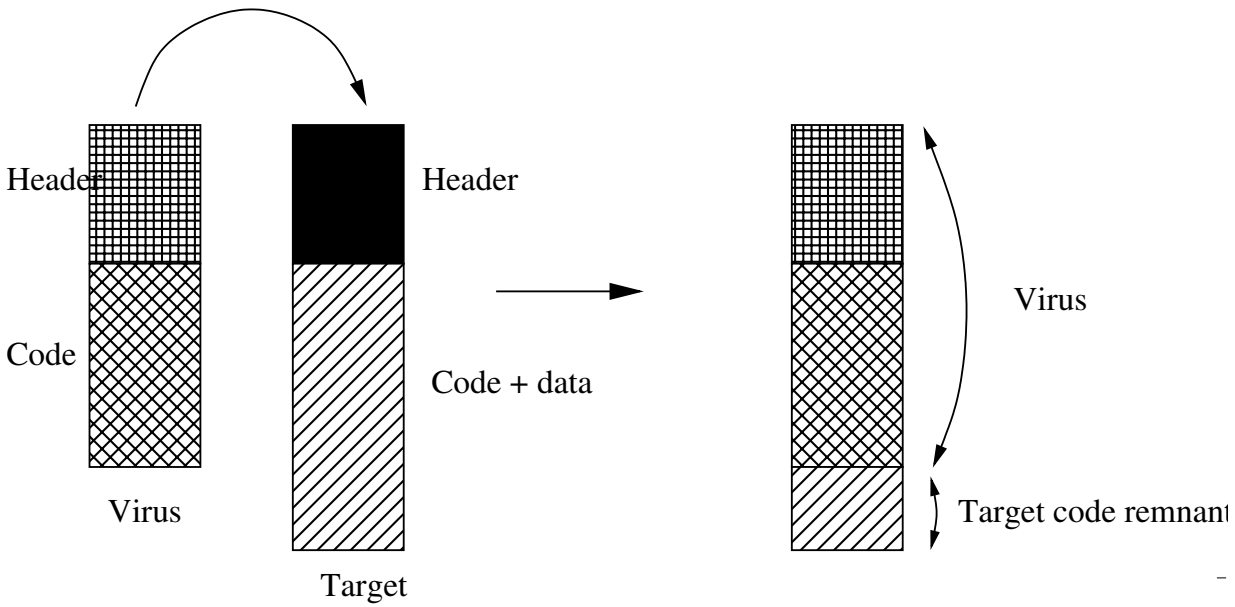
Overwriting Viruses

Three overwriting cases are possible:

❶ The virus overwrites the initial portion of the target code.

❷ The virus overwrites the middle or terminal portion of the target code.

❸ The virus simply replaces the target code with its own code.



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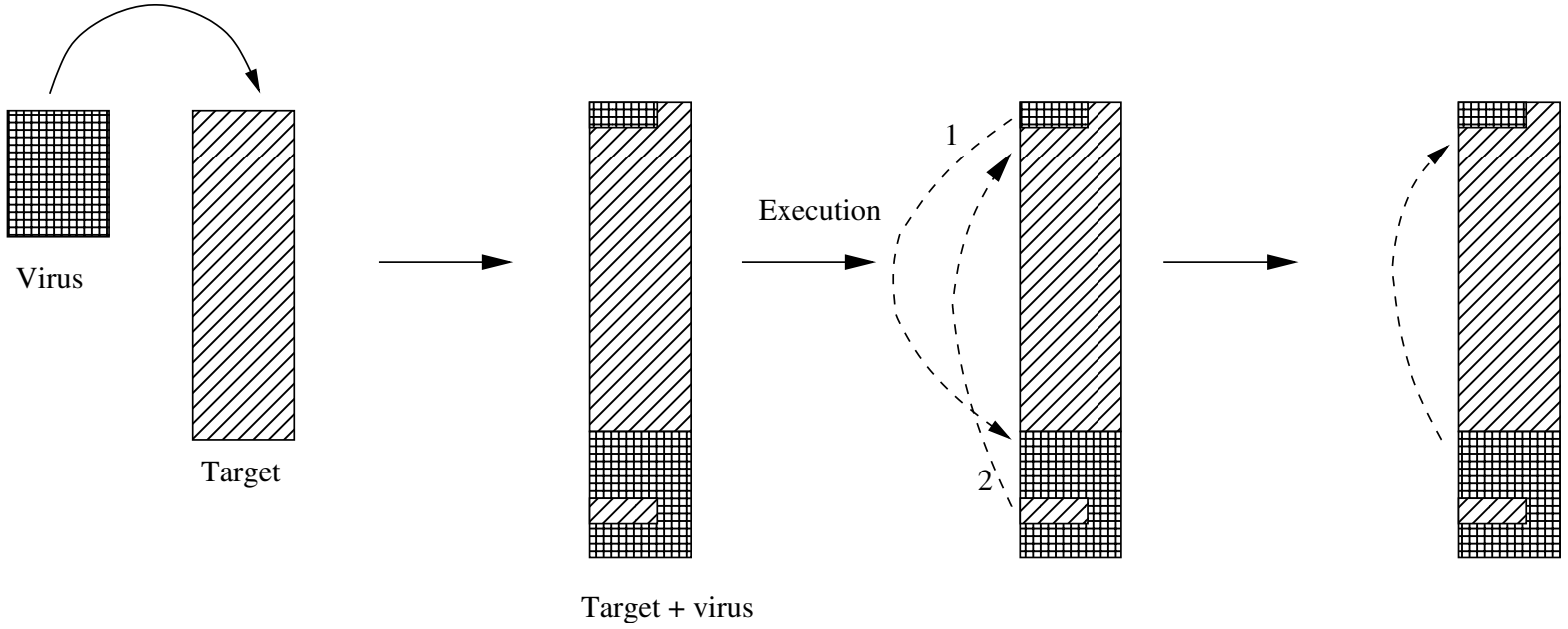
Prepending and Appending Viruses

Viruses using this method attach their code to the target. Two cases:

- *At the beginning of the target code (prepending).* This is complex because the header must be modified:
  - File length and possibly code sections.
  - Address recalculation.
  - Additionally, a *jump* to the beginning of the target code must be added.

Prepending and Appending Viruses (2/2)

- *At the end of the target code (appending).*
  - The beginning of the target code must be replaced with a jump to the viral code.
  - The replaced code must be executed at the end of the viral code.
  - Followed by a jump back to the target code.



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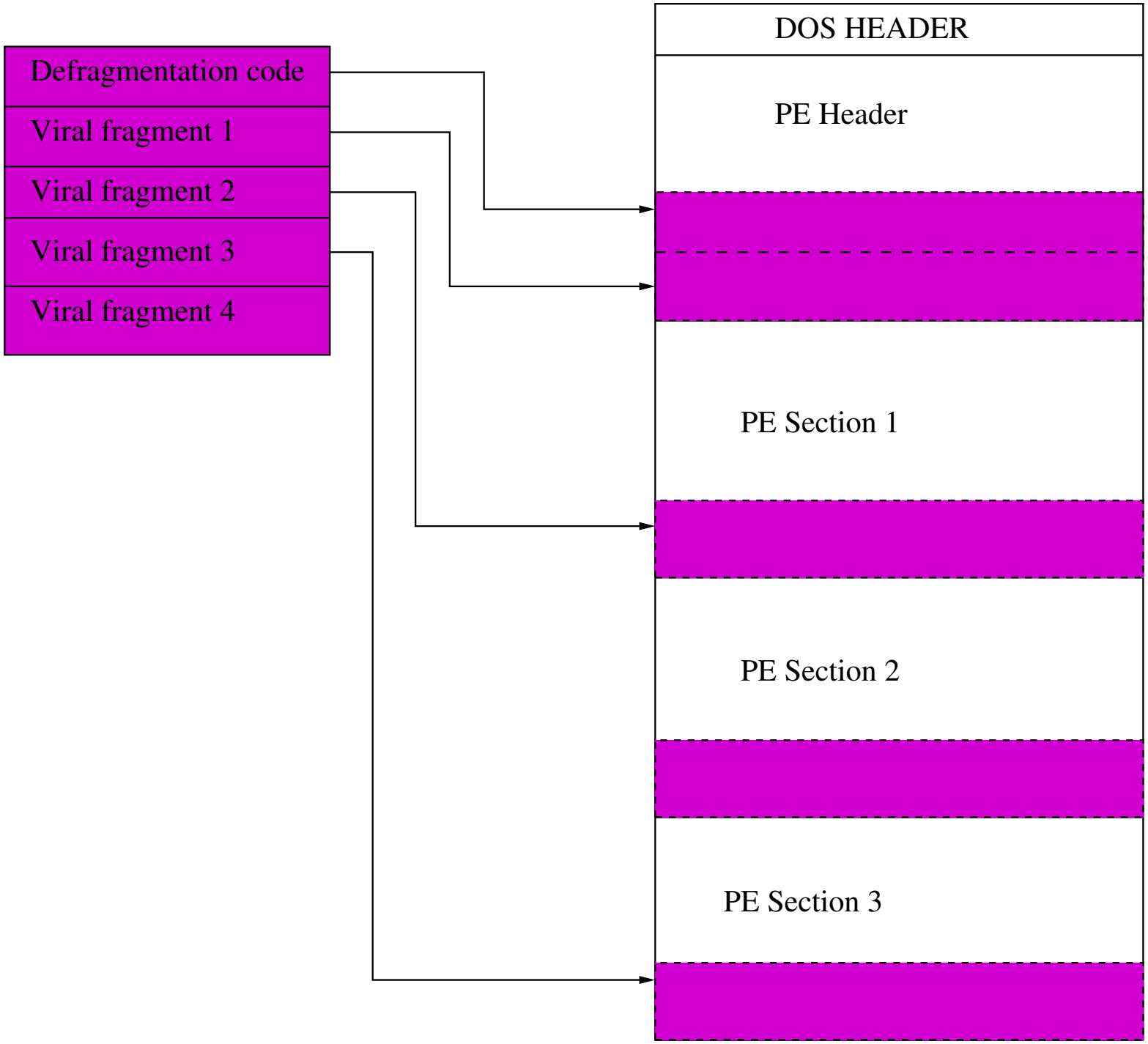
Cavity Viruses

These viruses primarily exploit the PE (Portable Executable) format of 32-bit Windows executables. This header enables during memory mapping:

- providing adequate information for memory installation (establishing a memory image);
- allowing optimal sharing between multiple processes of EXE and DLL files.

This format lends itself particularly well to virus writing.


Defragmentation code
Viral fragment 1
Viral fragment 2
Viral fragment 3
Viral fragment 4
DOS HEADER
PE Header
PE Section 1
PE Section 2
PE Section 3

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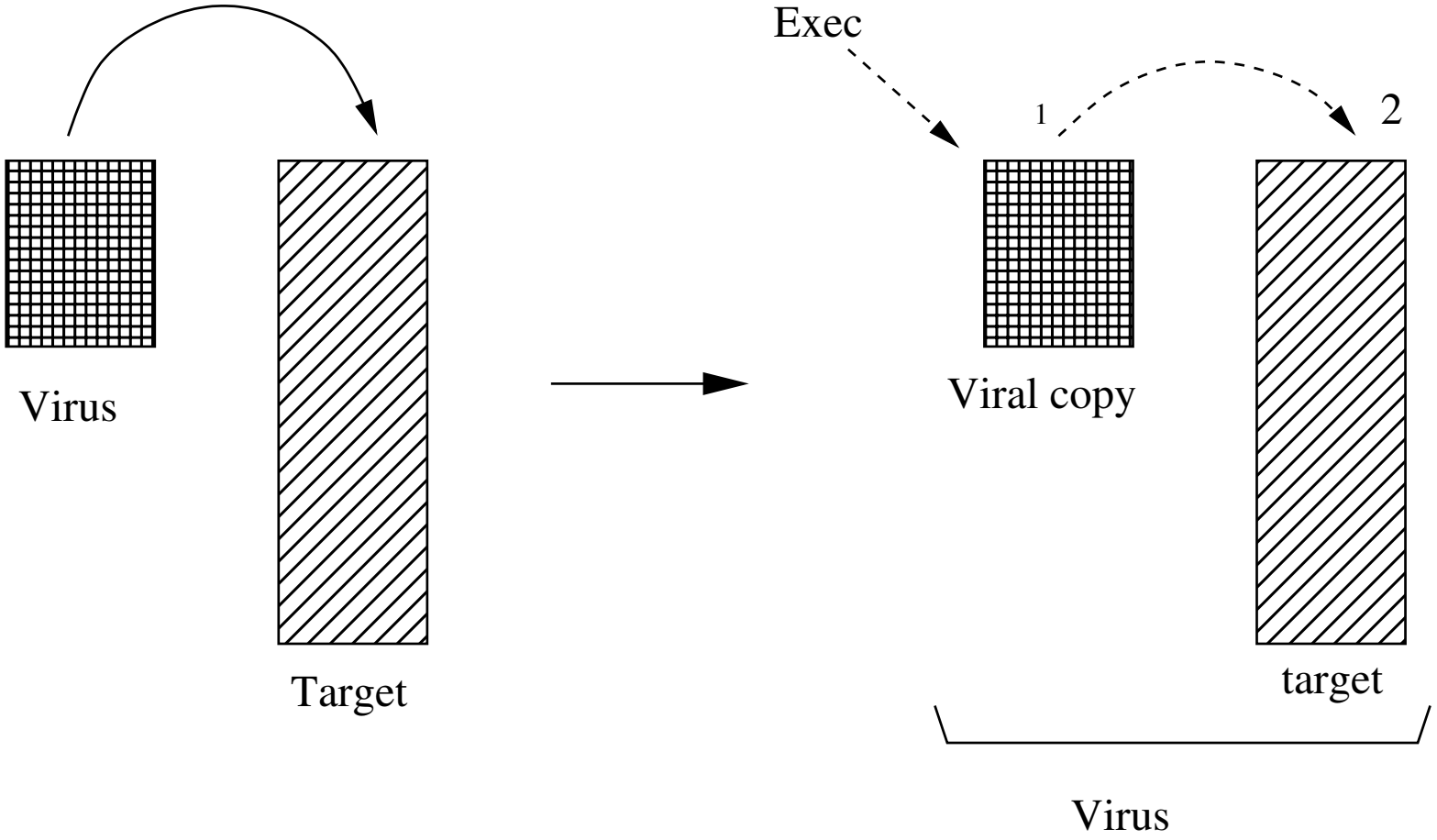
Companion Viruses

- Least known method.
- The target code is not altered.
- The viral code takes the name of the original program and executes the target code using the `exec` or `system` command.



Companion Viruses - Linux Example

- On Linux, the virus modifies the PATH by prepending the current directory ".".
- In the current directory, a copy of the virus is made under the name `ls`.
- The virus code is modified so that it launches the system `ls` command at the end of its execution.

A similar approach can be used on Windows.

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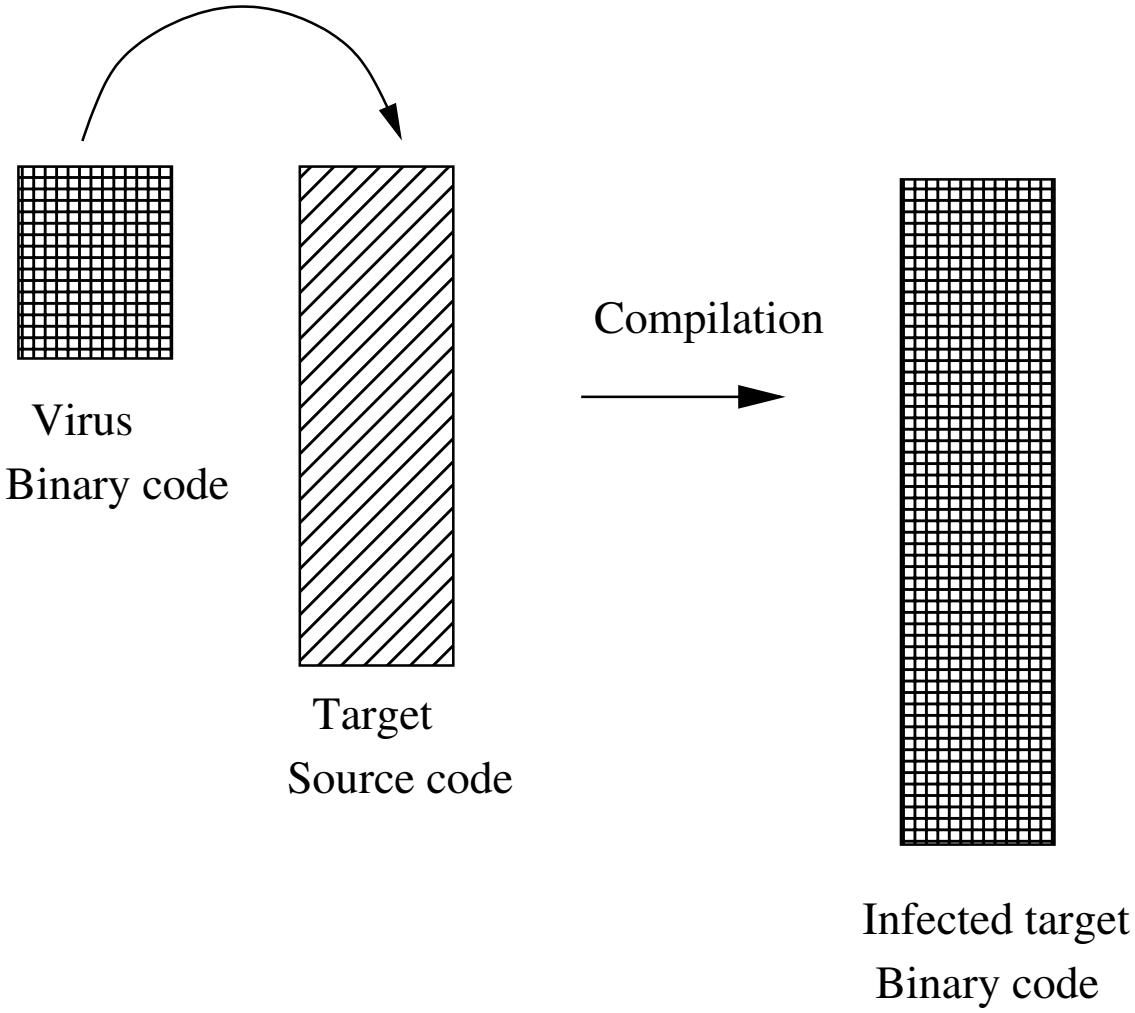
Source Code Viruses

- The target here is the source code of a C program.
- The infected program must be recompiled to produce a valid executable.
- A quick analysis of the infected code could reveal the presence of the virus, so the code must be duplicated in a more sophisticated manner.
- The advantage of source code infection is that the resulting binary is completely homogeneous.

Source Code Viruses



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Anti-Antivirus Techniques

- Malware evolves to counter and defeat protections implemented by antivirus software or firewalls.
- *Stealth.* Used to deceive the user, the system, and protection software by making them believe there is no malicious code present.
- *Polymorphism.* Since antivirus software largely works by searching for viral signatures, the goal of polymorphism is to vary the viral code from copy to copy.

Anti-Antivirus Techniques: Polymorphism

The two main techniques are as follows:

- Code rewriting using equivalent code. For example in C language:

  ```
  if( flag ) infection(); else payload();
  ```

  can be rewritten as follows:

  ```
  (flag) ?  infection() :  payload();
  ```

- Using basic encryption techniques on all or part of the virus or worm.

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Virus Classification

Viruses and worms can be classified according to different criteria:

- by target format: executable viruses or document viruses.
- by target component: boot sector viruses, device driver viruses,. . . .
- by programming language: assembly viruses, source code viruses, script viruses.
- by behavior: armored viruses, slow or fast viruses, retroviruses, resident viruses, polymorphic or stealth viruses,. . .

Document Viruses or Macro Viruses

We will adopt the following definition:

Definition (Document Viruses)

A document virus is:

- *Viral code contained in a data file, non-executable.*

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A document virus is:

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- *It is activated by an interpreter natively contained in the application associated with the file format (determined by its extension).*

Document Viruses or Macro Viruses

We will adopt the following definition:

Definition (Document Viruses)

A document virus is:

- *Viral code contained in a data file, non-executable.*
- *It is activated by an interpreter natively contained in the application associated with the file format (determined by its extension).*
- *The malicious code activation is performed:*
  - *either by a feature provided in the application,*
  - *or by exploiting an internal vulnerability of the application.*

Document Viruses: Risks Associated with Known Formats

Format	File Extensions	Risk	Type
WSH Script	VBS,JS,VBE,JSE,WSF,WSH	1	text
Word	DOC,DOT,WBK,DOCHTML	2,3,	binary
Excel	XLS,XL?,SLK,XLSHTML	2,3	binary
Powerpoint	PPT, POT,PPS, PPA,PWZ,PPTHTML, POTHTML	2,3	binary
Access	MDB,MD?,MA?,MDBHTML	1	binary
RTF	TRTF	4	text
Shell Scrap	SHS	1	binary
HTML	HTML,HTM,...	2	text
XHTML	XHTML,XHT,...	2	text
Adobe Acrobat	PDF	2	text
PostScript	PS	1,2	text
tex,latex	TEX	?	text

BIOS Viruses - Boot Sector Viruses

BIOS Viruses:

- When the computer is powered on, the code contained in the BIOS is activated.
- Its role is to test the hardware environment and then pass control to the boot sector.
- A BIOS virus modifies the BIOS code.

Boot Sector Viruses:

- The viral code is placed in the boot sector of the hard drive.
- It executes before any other program, allowing it to disable antivirus protection before it runs.

Memory-Resident Viruses

This is a virus that is constantly running.

- A resident virus is a virus that, once executed, remains active: only shutting down the machine or a `kill` command can stop it.
- It uses various techniques to be re-executed after a machine shutdown (launch at startup, etc.).
- It can interfere with the system (bypass certain system interrupts, for example).

Virus Hoax

This is a kind of viral code that executes in the brain.

Definition (Virus Hoax)

A virus hoax is misinformation that uses social engineering techniques to induce users to produce effects equivalent to those of a virus or worm: propagation and offensive action.

Example: circulation of an email warning of an infection, solved by deleting a system file.

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Classification

Simple Worms:

- These are the ones that can legitimately be called worms.
- They generally exploit software and network vulnerabilities allowing program execution on a remote machine.

Macro Worms (virus-worm hybrids):

- Dissemination occurs through attachments containing infected office documents sent to users in the address book.

Email Worms:

- Propagation through attachments containing malicious code.
- The code is activated either by the user or by the email application.
- The most famous example is the ILOVEYOU worm.

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Key Takeaways

This chapter provides general information about malware:

- Understand the different types of malware.
- Distinguish between viruses and worms.
- Understand virus replication methods.
- Understand virus propagation methods.